Event Duration Monitoring - Storm Overflow Annual Return - 2024

Dataset documentation

Date: 27 March 2025

We are the Environment Agency. We protect and improve the environment.

We help people and wildlife adapt to climate change and reduce its impacts, including flooding, drought, sea level rise and coastal erosion.

We improve the quality of our water, land and air by tackling pollution. We work with businesses to help them comply with environmental regulations. A healthy and diverse environment enhances people's lives and contributes to economic growth.

We can’t do this alone. We work as part of the Defra group (Department for Environment, Food & Rural Affairs), with the rest of government, local councils, businesses, civil society groups and local communities to create a better place for people and wildlife.

Published by:

Environment Agency
Horizon House, Deanery Road,
Bristol BS1 5AH

www.gov.uk/environment-agency

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Contents

Dataset description

The dataset includes one Excel file (containing 10 separate spreadsheets, one for each WaSC with storm overflows in England) reporting how often and for how long monitored storm overflows discharged during 2024. WaSCs provide this regulatory return to the Environment Agency each year to fulfil their permitted conditions to discharge from these storm overflows under the Environmental Permitting Regulations.

A separate Excel file (and .pdf) summarises the key data from the 10 WaSC spreadsheets including; how many storm overflows were monitored, and the average spill duration and spill count per monitored overflow for each WaSC. Additional contextual information is provided. This includes information relating to the operability of the EDM devices and actions to resolve any issues encountered with the monitors within the year. It also includes categorisation of reasons for high spill frequency overflows and action taken/planned to resolve these. See Appendix A for these categories. The additional summary dataset has been produced by the Environment Agency to show key findings, variations and trends.

Update frequency

This dataset will be updated annually in March.

Related datasets

This dataset presents the performance of permitted (under Environmental Permitting Regulations) storm overflows. Further details of permitted storm overflows can be found in our dataset 'Consented Discharges to Controlled Waters with Conditions' (updated quarterly).

Common questions and known issues

The key performance measures are in the following columns:

- **Total Duration (hh:mm:ss) of all spills prior to processing through 12-24 hour counting method (column P)** - how many hours the storm overflow was measured to discharge to the environment during the reporting period in 2024.
- **Counted spills using 12-24 hour count method** (column Q) - how many occurrences the storm overflow was measured to discharge to the environment during the reporting period in 2024. The 12-24hr counting method ensures that very long continuous spills over multiple days are not counted as one spill. It is described in our guidance; 'Water companies: environmental permits for storm overflows and emergency overflows - Updated 13 September 2018' found here: https://www.gov.uk/government/publications/water-companies-environmental-permits-for-storm-overflows-and-emergency-overflows/water-companies-environmental-permits-for-storm-overflows-and-emergency-overflows
- **% of reporting period EDM operational** (column T) - the percentage of the reporting period that the monitor was functioning and could reliably record discharges if one occurred.
- We do our best to avoid quality problems, and this dataset reflects the data we hold. However, this dataset may contain errors.

Dataset content

EDM 2024 Storm Overflow Annual Return - all water and sewerage companies.xlsx

This file contains ten separate spreadsheets, one for each data return from each WaSC:

- Anglian Water (AWS)
- Dŵr Cymru Welsh Water (DCWW) (in England)
- Northumbrian Water (NW)
- Severn Trent Water (SvT)
- South West Water (SWW)
- Southern Water (SW)
- Thames Water (TW)
- United Utilities (UU)
- Wessex Water (WSSX)
- Yorkshire Water (YWS)

There is also a summary spreadsheet and pdf that contain key summary data:

EDM 2024 Storm Overflow Annual Return - summary data.xlsx

EDM 2024 Storm Overflow Annual Return - summary data.pdf

Fields applying to the 10 Water and Sewerage Company datasets

The following fields apply to the 10 WaSC datasets (**EDM 2024 Storm Overflow Annual Return - all water and sewerage companies.xlsx**):

Unique ID (Column A)

- A unique reference for each discharge activity. Enables the longer term tracking of data where other references may change.

Water Company Name (Column B)

- The name of the Water and Sewerage Company (WaSC) that is permitted to operate the storm overflow and who made the data return to the Environment Agency

Site Name (EA Consents Database) (Column C)

- The permitted site name of the storm overflow held by the Environment Agency
- This name may vary from the Consents Database when agreed with the Environment Agency or a more appropriate reference exists.

Site Name (WaSC operational) [optional] (Column D)

- Site name used by the WaSC for operational reasons

EA Permit Reference (EA Consents Database) (Column E)

- The permit reference held by the Environment Agency
- This reference may vary from the Consents Database when agreed with the Environment Agency or a more appropriate reference exists.

WaSC Supplementary Permit Ref. [optional] (Column F)

- WaSC reference number/name for the overflow

Activity Reference on Permit (Column G)

- Identifies which permitted discharge is being measured when more than one discharge is referenced on the permit

Storm Discharge Asset Type (Column H)

- Identifies the type of overflow
- See Appendix A for the data dropdown options appropriate to this column

Outlet Discharge NGR (EA Consents Database) (Column I)

- Location of discharge point to the environment
- Note the overflow & EDM device may be located further up the sewer network
- This NGR may vary from the Consents Database when agreed with the Environment Agency or a more appropriate NGR exists.

WFD Waterbody ID (Cycle 3) (discharge outlet) (Column J)

- Identification number of Water Framework Directive (WFD) waterbody at the discharge point (as per Cycle 3)

WFD Waterbody Catchment Name (Cycle 3) (discharge outlet) (Column K)

- Name of WFD waterbody catchment at the discharge point (as per Cycle 3)

Receiving Water / Environment (common name) (EA Consents Database) (Column L)

- Name of the receiving water at discharge point, as recorded on the permit
- This name may vary from the Consents Database when agreed with the Environment Agency or a more appropriate reference exists.

Shellfish Water(s) (only populate for storm overflow with a Shellfish Water EDM requirement) (Column M)

- Name of the designated shellfishery

Bathing Water(s) (only populate for storm overflow with a Bathing Water EDM requirement) (Column N)

- Name of the designated bathing water

Treatment Method on Permit (over & above storm tank settlement/screening) (Column O)

- Whether the storm overflow discharge is subject to any treatment method which is detailed on the permit.
- See Appendix A for the data dropdown options appropriate to this column

Total Duration (hh:mm:ss) all spills prior to processing through 12-24h count method (Column P)

- How many hours the storm overflow was measured to discharge to the environment in 2024

Counted spills using 12-24h count method (Column Q)

- How many times the storm overflow was measured to discharge to the environment in 2024
- The 12-24hr counting method ensures that very long continuous spills over multiple days are not counted as one spill. It is described in our guidance; 'Water companies: environmental permits for storm overflows and emergency overflows - Updated 13 September 2018' found here: https://www.gov.uk/government/publications/water-companies-environmental-permits-for-storm-overflows-and-emergency-overflows/water-companies-environmental-permits-for-storm-overflows-and-emergency-overflows

Long-term average spill count (Column R)

- Average spill count based on all existing verifiable and validated EDM data at this storm overflow

Data start – Calendar Year (Column S)

- Calendar year when EDM data was first recorded. This provides context for the long-term average spill count (column R)

EDM Operation - % of reporting period EDM operational (Column T)

- The percentage of the reporting period that the monitor was functioning and could reliably record discharges if one occurred.

EDM Operation - Reporting % - Primary Reason <90% (Column U)

- Primary reason why the EDM may not have been operational ≥90% of the reporting period
- Category must be selected when period of EDM operation (column T) is less than 90%. Optional to complete this column when period of EDM operation is 90% or over
- Data represents best information as held by the WaSC at time of submission of the Annual Return
- See Appendix A for the data dropdown options appropriate to this column

EDM Operation - Action taken / planned - Status & timeframe (Column V)

- Indicates whether action has already been taken / is planned to be taken, and in which month; or whether there is an ongoing investigation to identify the appropriate action
- See Appendix A for the data dropdown options appropriate to this column

High Spill Frequency - Operational Review - Single year reason (Column W)

- Core reason for spill count (column Q) within single reporting year
- Category must be completed for all storm overflows on the return - A reason other than 'High spill threshold not met' should be selected when spill frequency in column Q exceeds the single year EDM Annual Return high spill frequency threshold currently >60
- Captures information about whether spill frequency is above the single year EDM Annual Return high spill frequency threshold.
- Data represents best information as held by the WaSC at time of submission of the Annual Return
- See Appendix A for the data dropdown options appropriate to this column

High Spill Frequency - Operational Review - Long-term average reason (Column X)

- Core reason for long-term average spill count (column R)
- Category must be selected when spill frequency in column R exceeds either of the long-term average EDM Annual Return high spill frequency thresholds; currently >50 with 2 calendar years existing data in column S OR column R >40 with 3 or more calendar years existing data in column S
- Captures information on why spill frequency is above EDM Annual Return high spill frequency threshold
- See Appendix A for the data dropdown options appropriate to this column

Investigation activity for the reporting period (Column Y)

- Captures information on investigations that have taken place in relation to the storm overflow during the specified Annual Return reporting period (calendar year)
- An investigation is defined as all work that might be carried out up until improvement work begins on site
- See Appendix A for the data dropdown options appropriate to this column

Improvement activity for the reporting period (Column Y)

- Captures information on improvement works that have taken place in relation to the storm overflow during the specified Annual Return reporting period (calendar year)
- An improvement is defined as all works involving on site actions
- Improvements are in the context of spill reduction activity rather than operational activity like cleaning a screen which would not be deemed as an improvement
- See Appendix A for the data dropdown options appropriate to this column

UWWTR (SOAF) investigation status (Column AA)

- Captures the status of SOAF investigations at the time of EDM Storm Overflow Annual Return submission
- See Appendix A for the data dropdown options appropriate to this column

Old format Unique ID - pre-2024 Annual Return (Column AB)

- Captures previously used Unique ID for cross-reference purposes
- Superseded by new format Unique ID now found in column A
- Entry required for storm overflows where an old format Unique ID was assigned in the 2023 EDM Annual Return or the 2024 EDM Bathing Season return.

Fields applying to the Summary Dataset

The following fields apply to the summary dataset (**EDM 2024 Storm Overflow Annual Return - summary data.xlsx** and **EDM 2024 Storm Overflow Annual Return - summary data.pdf**):

Table 1: 2024 EDM Headlines

Total no. storm overflows listed in the annual return in 2024

- The number of storm overflows included by Water & Sewerage Companies (WaSCs) in the 2024 Event Duration Monitoring Annual Return
- WaSCs should have listed all their Storm Overflows, regardless of whether they are currently permitted or not, or whether they have EDM commissioned or not

Total no. active storm overflows listed in the annual return in 2024

- The number of active storm overflows included by Water & Sewerage Companies (WaSCs) in the 2024 Event Duration Monitoring Annual Return
- This is the number of overflows that were able to operate during the reporting year and excludes overflows that were revoked or surrendered at or before the start of the year.
- Total number of overflows minus overflows which Water companies would not be expected to report EDM data for.

Total no. storm overflows with EDM commissioned

- The number of storm overflows each WaSC declared had EDM commissioned (reliable data can be expected) by the end of the reporting year

% active storm overflows listed with EDM commissioned

- Percentage of active storm overflows listed that had EDM commissioned (reliable data can be expected) by the end of the reporting year
- Total number of storm overflows reported to have a data start point (column S) divided by the total number of 'active' storm overflows listed in the Annual Return by that WaSC
- The water company average for this row is calculated using the total number of 'active' overflows for all companies rather than as an average of the company averages.

Total no. storm overflows with spill data in 2024

- The number of storm overflows in the EDM annual return which have spill count data reported (column Q)
- Some storm overflows may have an EDM commissioned but no reliable spill data is provided. This is either due to 0% EDM Operation during the reporting year, or where the EDM was working but a WaSC has not reported the spill data in the annual return.

Average no. spills per storm overflow with spill data in 2024

- The average number of spills that were counted (12/24h count method) per overflow with spill data.
- Total number of spill events reported by a WaSC divided by the total number of storm overflows with spill data reported by the WaSC
- The water company average for this row is calculated using the total number of spill events reported by all WaSC's divided by the total number of storm overflows with spill data reported by all WaSC's

Average duration (hrs) per monitored spill event in 2024

- The average duration of a spill per monitored overflow
- Sum duration of spill hours reported by a WaSC divided by the total number of spill events reported by that WaSC
- The water company average for this row is calculated using the sum duration of spill hours reported by all WaSC's divided by the total number of spill events reported by all WaSC's

Table 2: 2024 EDM Summary Statistics

Total no. number of spill events in 2024

- The total number of spill events monitored during 2024

Average no. spills per storm overflow with spill data in 2024

- See Table 1, metric (row) 6

Total duration (hrs) of monitored spill events in 2024

- Total hours monitored storm overflows discharged during the year
- Note that multiple storm overflows are likely to be discharging at the same time in response to rainfall and/or snowmelt

Average duration (hrs) per monitored spill event in 2024

- See Table 1, metric (row) 7

% storm overflows spilled ≤10 times in 2024

- Percentage of monitored storm overflows that recorded 10 spills or less in the reporting period
- No. overflows ≤10 spills (column Q) expressed as a percentage of the total overflows with spill data reported by the WaSC
- The water company average for this row is calculated using the total number of overflows with ≤10 spills (column Q) expressed as a percentage of the total overflows with spill data reported by all WaSC's

Percentage time operating [spilling] during 2024 per overflow (average)

- Percentage of the year that the average monitored overflow operated
- Average duration of a monitored overflow multiplied by the average number of spills per storm overflow with spill data reported by the WaSC. Then multiplied by 24h x (365 days equal to 8760 hours in a standard year or 366 days equal to 8784 hours in a leap year).

Table 3: 2024 EDM Device Operation

Total no. storm overflows with EDM Operation data

- The number of storm overflows with data showing whether the monitor was operational during the period (e.g. from 0% to 100%) (column T)

% storm overflows with EDM Operation data provided where expected

- Percentage of storm overflows with a data start date (Column S) and % EDM Operation data is provided (Column T)

% storm overflows with 0% EDM Operation during reporting period

- Percentage of monitored storm overflows that did not return any reliable spill count data during the reporting period

% storm overflows with ≥90% EDM Operation during reporting period

- Percentage of monitored storm overflows that provided reliable data for 90% (or over) of the reporting period

% storm overflows with <90% EDM Operation during reporting period

- Percentage of monitored storm overflows that provided reliable data for less than 90% of the reporting period

% of those with <90% operability with reason provided

- Percentage of monitored storm overflows that provided reliable data for less than 90% of the reporting period, with the primary reason included within the dataset (column U)

Table 4: 2024 Storm Overflow Spill Performance

% with EDM installed & provided count data - with 0 spill count (did not spill)

- Percentage of monitored storm overflows that recorded zero spills during the reporting period

% storm overflows with spill data - recorded ≥1 spill count

- Percentage of monitored storm overflows that recorded one or more spills during the reporting period

% recorded 5 spills or less

- Percentage of monitored storm overflows that recorded five spills or less in the reporting period

% recorded 10 spills or less

- Percentage of monitored storm overflows that recorded 10 spills or less in the reporting period

% recorded >10 spills

- Percentage of monitored storm overflows that recorded 11 or more spills in the reporting period

% recorded 20 spills or more

- Percentage of monitored storm overflows that recorded 20 spills or more in the reporting period

% recorded 40 spills or more

- Percentage of monitored storm overflows that recorded 40 spills or more in the reporting period

% recorded 60 spills or more

- Percentage of monitored storm overflows that recorded 60 spills or more in the reporting period

% recorded 100 spills or more

- Percentage of monitored storm overflows that recorded 100 spills or more in the reporting period

% recorded 200 spills or more

- Percentage of monitored storm overflows that recorded 200 spills or more in the reporting period

Table 5: 2024 Storm Overflow Spill Reasons

(Based on best available information held by the WaSC at the time of Annual Return data submission)

No. monitored storm overflows that spilled >60 in reporting year

- The number of monitored overflows with greater than 60 spills in the reporting period

Of those that spilled over SOAF thresholds of >60x in reporting year, % with a reason provided

- % of overflows with greater than 60 spills in the reporting period that had a primary reason provided in column W
- The threshold of greater than 60 spills is based on spill frequency investigation triggers in the Storm Overflow Assessment Framework version 1

Of those that spilt over SOAF thresholds of >60x in one year, what % due to exceptional weather throughout the year

- % of overflows with greater than 60 spills in the reporting period with primary reason attributed to exceptional weather
- Exceptional weather does not refer to individual rainfall events, but rather the rainfall across the reporting year.
- Two datasets can be used to determine whether rainfall in the reporting year was "exceptional" or not (over & above typical rainfall) – (1) Environment Agency water situation reports or (2) local rainfall records.
- If rainfall was exceptional and deemed the primary reason for a high spill count then this is indicated by the WaSC in the annual return (Column W)

Of those that spilt over SOAF thresholds of >60x in one year, what % due to other operational reasons (incl. asset maintenance)?

- % of overflows with greater than 60 spills in the reporting period with primary reason attributed to other operational reasons, including asset maintenance
- Operational reasons (incl. asset maintenance) are where the asset (storm overflow) and potentially parts of the upstream & downstream sewer network have not operated as designed/expected.
- If asset maintenance is deemed the primary reason for high spill count this is indicated by the WaSC in the annual return (column W). The different operational reason categories are listed in Appendix A of this dataset README guide

Of those that spilt over SOAF thresholds of >60x in one year, what % due to hydraulic capacity reasons?

- % of overflows greater than 60 spills in the reporting period with primary reason attributed to hydraulic capacity
- If the reason for a high spilling storm overflow (over 60 times per year) is neither “exceptional rainfall” nor “asset maintenance” then the reason is classified under the “Hydraulic capacity” category.
- This indicates there is insufficient capacity (conveyance or storage) in the sewer network to cope with the wastewater flow plus typical rainfall entering the sewer network.

Of those that spilt over SOAF thresholds of >60 in one year, what % N/A - ongoing investigation for primary reasons

- % of overflows greater than 60 spills in the reporting period with an investigation ongoing, which has not yet indicated the suspected primary reason by the time of Annual Return submission in February.
- This category should be used by exception only; for example where high spill frequency is crossed late in the reporting year and a WaSCs has had limited opportunity to determine an initial primary reason for spill count

Table 6: 2024 EDM Storm Overflow Annual Return Data Entry

Total no. storm overflows listed in the annual return in 2024

- See Table 1, metric (row) 1

Total no. storm overflows with EDM commissioned

- See Table 1, metric (row) 3

% active storm overflows listed with EDM commissioned

- See Table 1, metric (row) 4

Total no. storm overflows with EDM device operability data

- See Table 3, metric (row) 1

Total no. overflows with spill data

- See Table 1, metric (row) 5

% overflows listed with spill data

- Percentage of storm overflows listed in the return that have spill data reported
- Reasons for less than 100% include no EDM yet commissioned; EDM commissioned but 0% operability (therefore no spill data); or where the EDM was working but a WaSC has not reported the spill data in the annual return

% overflows with <90% operability where valid reason provided

- See Table 3, metric (row) 6

% of overflows >60 spills in one year with a reason provided

- See Table 5, metric (row) 2

Appendix A

Appendix A shows the drop-down categories applicable to certain columns on the 2024 EDM Annual Return.

Column H: Storm Discharge Asset Type

SO on sewer network

- Storm overflow on sewer network. This can be direct or associated with a network storm tank
- Not associated with pumping station

Storm discharge at pumping station

- Storm overflow on network at pumping station

Inlet SO at WwTW

- Storm overflow (direct to the environment), permitted as part of a WwTW permit
- Exception is storm tank - see note below

Storm tank at WwTW

- Storm overflow via a storm tank, permitted as part of a WwTW permit
- Note this can be physically remote from the WwTW

Other storm discharge asset type

- Any storm overflow that does not clearly align with the four categories above
- These may be used to develop future drop-down categories

(As the five categories above but with "- with treatment" added)

- As above, but indicates any storm spills are subject to treatment (over & above storm tank settlement/screening) before reaching the water environment

Column O: Treatment method on permit (over & above storm tank settlement/screening)

Not applicable

- No treatment method applies or is not recognised in the permit

UV (Ultraviolet)

Chemical disinfection

Membrane filtration

Reed bed

Facultative ponds

Column U: EDM Operation - Reporting Percentage - Primary Reason <90%

Access – Unable to retrieve data from non-telemetry data logger

- Not to be used for delayed access to fix existing fault
- When access causes inability to retrieve data in part/full; e.g. landowner permission
- Consider core reason – if caused by poor installation or design then select “Installation set-up/design issue” category instead
- Other examples include inability to retrieve data due to highway access/ parked vehicles / unsafe access conditions

Capital / maintenance works affect EDM operation

- Where planned works affect EDM operation.
- Consider core reason – if caused by poor installation or design then select “Installation set-up/design issue” category instead.
- Other examples include where false data provided due to chambers not being cleared/maintained

Comms failure / issue

- Any part of communication failure; e.g. intermittent signal or antenna damage.
- Consider core reason – if caused by poor installation or design then select “Installation set-up/design issue” category instead. Similar if comms loss caused by power failure – select “Power failure / issue” instead
- Other examples include third party damage / intermittent issues caused by vehicles parked over manholes / poor signal strength in remote locations / loss of third party network provision problem (particularly after storms)

Installation set-up / design issue

- When installation or design (e.g. choice of location) affects EDM operability.
- e.g. Original design location cannot distinguish between two overflows & original EDM needs to be relocated to more representative point
- e.g. Original design location affected by river ingress & EDM requires relocation
- e.g. Alternative monitor type required
- Other examples include EDM on storm tanks recording high storm tank volume rather than a spill to the environment / EDM device not set up to read at required interval (e.g. 15

minutes instead of 2 minutes) / interference from other ultrasonic monitors / physical structure of storm overflow (e.g. uneven bench or shallow chambers) causing false spill recordings due to echo bounce

Power failure / issue

- Any part of power failure; e.g. loss of mains supply / battery fault
- Consider core reason – if caused by cutting through cable then select "Capital / maintenance works" category instead
- Other examples include unable to replace battery due to unsafe access / third party vandalism to power supply

Sensor failure / issue

- Any part of sensor failure;e.g. water ingress to connection between sensor & logger.
- Consider core reason – if water ingress caused by poor choice of location for installation & requires adjustment then select "Installation set-up/design issue" category instead.
- Other examples include third party interference (e.g. vegetation; unflushables; vandalism) / extreme temperature changes causing drift in readings

Telemetry or data archiving failure / issue

- Any part of telemetry failure; e.g. dial-in or storage/sending of data problems.
- Consider core reason – if archiving problem caused by power fault or incorrect installation then use "Power failure / issue" or "Installation set-up/design issue" categories respectively. Similar if data transfer caused by intermittent signal – select "Comms failure / issue" instead
- Other examples include when data found to be associated with the wrong site / software issues / outstation failure

No longer operational as an overflow – permit revoked or to be revoked

- If asset operational as an overflow at some point in the reporting year then the EDM must have been operational & recording potential spills before the overflow was no longer operational. These data must be reported.
- Also includes where permit condition still existed at some point during the reporting year - asset must still be reported on the EDM storm overflow Annual Return
- Also includes asset is no longer operational therefore data not available or not representative of any discharge to the environment

Column V: EDM Operation - Action taken / planned - Status & timeframe

Scheduled

- Appropriate operational action to address <90% EDM operation is planned

Resolved - (mmm)

- Appropriate operational action has already been taken within the reporting year & primary issue affecting EDM operation resolved

Resolved - (mmm) after reporting year

- Appropriate operational action has been taken & primary issue affecting EDM operation resolved between the end of the reporting year & submission of the regulatory EDM Annual Return by the end of February

N/A - Ongoing investigation

- Appropriate action not yet identified
- Also includes appropriate action identified but not yet scheduled / previous resolution but further investigation is now required

Overflow no longer spilling to environment - from (mmm)

- Use when selection in column U is "No longer operational as an overflow - permit revoked or to be revoked".
- Month represents month from which spills to the environment were no longer possible (e.g. overflow sealed off in network)

Column W: High Spill Frequency – Operational Review – Single year reason

Performance - Partial / no capacity due to blockage or restriction - maintenance issue - (Asset maintenance)

- Spill frequency primarily caused by maintenance issue
- e.g. roots causing channel restriction
- e.g. blocked screens causing premature spills

Performance - Sewer collapse (partial / full) - infrastructure issue - (Asset maintenance)

- Spill frequency primarily caused by infrastructure issue
- e.g. partial collapse of sewer

Performance - Groundwater inundation - (Asset maintenance)

- Groundwater inundation is primary reason for spills
- e.g. GW inundation in chalk catchment causing high spill frequency
- Groundwater inundation can be caused by high water table or flooding in the catchment

Performance – Infiltration - (Asset maintenance)

- High spill frequency caused by infiltration. Requires investment to infrastructure to resolve Groundwater or surface flows enter the sewerage network via defects such as loose joints or cracks. These defects can be in public or private sewers.
- Infiltration can be caused by misconnection of surface water pipes or large areas of impermeable ground being connected to the sewerage network

Performance - Asset power failure - (Asset maintenance)

- E.g., frequent power supply failure caused high spill frequency e.g. fuse blown in asset and phase failure

Performance - Pump failure / issue - (Asset maintenance)

- Spill frequency primarily caused by pump failure / premature spills e.g., pumping station struggles to maintain PFF & review of rising main design required
- Also includes where pump capacity is inhibited by unflushable items and other things that shouldn't be in the sewerage network

Performance - Other maintenance / capital works - (Asset maintenance)

- Spill frequency primarily caused by works e.g., jetting
- Other examples include grit build up / air locking of rising main

Performance - Asset configuration (e.g. PS/rising main/storm tanks) - (Asset maintenance)

- Spill frequency primarily caused by inappropriate asset configuration
- e.g. inlet design causing premature spills & requires further investigation Other examples include weir height reduction over time causing premature spills / interference from screens

Data collection - EDM non-representative location - (Asset maintenance)

- E.g. EDM records multiple discharge points & cannot distinguish spill counts between the two.
- Requires EDM relocation
- Other examples include where EDM believed not to be recording only spills to environment (e.g. spills to balancing tank; high storm tank volume) / EDM affected by external noise / multiple monitors in network recording same spills to the environment

Data collection - Tidal / river inundation - (Asset maintenance)

- EDM spill frequency data quality affected by tidal or river inundation
- e.g. tidal cycle causes levels to reverse flows in outfall pipe-

Data collection - Confirmed exceptional weather - (Not asset maintenance)

- High spill frequency caused by exceptional rainfall events, over & above typical rainfall

Not asset maintenance - hydraulic capacity - (Not asset maintenance)

- Spill frequency primarily caused by hydraulic capacity issue rather than something that can be fixed operationally
- This indicates there is insufficient capacity (conveyance or storage) in the sewer network to cope with the wastewater flow plus typical rainfall entering the sewer network.
- Also includes where currently no clear evidence that the majority of spills were due to asset maintenance issues / assets where verified hydraulic model shows >40 spills are due to hydraulic overload / sites already part of ongoing SOAF investigations

N/A - Ongoing investigation – (Not asset maintenance)

- When investigation into high spill frequency has not yet indicated the suspected primary reason by the time of the regulatory EDM Annual Return submission at the end of February.

High spill threshold not met

- The single year EDM Annual Return high spill threshold of greater than 60 spills has not been exceeded.

Column X: High Spill Frequency – Operational Review – Long-term average reason

Performance - Partial / no capacity due to blockage or restriction - maintenance issue - (Asset maintenance)

- Spill frequency primarily caused by maintenance issue
- e.g. roots causing channel restriction
- e.g. blocked screens causing premature spills

Performance - Sewer collapse (partial / full) - infrastructure issue - (Asset maintenance)

- Spill frequency primarily caused by infrastructure issue
- e.g. partial collapse of sewer

Performance - Groundwater inundation - (Asset maintenance)

- Groundwater inundation is primary reason for spills
- e.g. GW inundation in chalk catchment causing high spill frequency
- Groundwater inundation can be caused by high water table or flooding in the catchment

Performance – Infiltration - (Asset maintenance)

- High spill frequency caused by infiltration. Requires investment to infrastructure to resolve
- Groundwater or surface flows enter the sewerage network via defects such as loose joints or cracks. These defects can be in public or private sewers.
- Infiltration can be caused by misconnection of surface water pipes or large areas of impermeable ground being connected to the sewerage network

Performance - Asset power failure - (Asset maintenance)

- E.g., frequent power supply failure caused high spill frequency e.g. fuse blown in asset and phase failure

Performance - Pump failure / issue - (Asset maintenance)

- Spill frequency primarily caused by pump failure / premature spills e.g., pumping station struggles to maintain PFF & review of rising main design required
- Also includes where pump capacity is inhibited by unflushable items and other things that shouldn't be in the sewerage network

Performance - Other maintenance / capital works - (Asset maintenance)

- Spill frequency primarily caused by works e.g., jetting
- Other examples include grit build up / air locking of rising main

Performance - Asset configuration (e.g. PS/rising main/storm tanks) - (Asset maintenance)

- Spill frequency primarily caused by inappropriate asset configuration
- e.g. inlet design causing premature spills & requires further investigation Other examples include weir height reduction over time causing premature spills / interference from screens

Data collection - EDM non-representative location - (Asset maintenance)

- E.g. EDM records multiple discharge points & cannot distinguish spill counts between the two.
- Requires EDM relocation
- Other examples include where EDM believed not to be recording only spills to environment (e.g. spills to balancing tank; high storm tank volume) / EDM affected by external noise / multiple monitors in network recording same spills to the environment

Data collection - Tidal / river inundation - (Asset maintenance)

- EDM spill frequency data quality affected by tidal or river inundation
- e.g. tidal cycle causes levels to reverse flows in outfall pipe

Data collection - Confirmed exceptional weather – Remaining spills not above SOAF threshold - (Not asset maintenance)

- High spill frequency caused by exceptional rainfall events, over & above typical rainfall

Not asset maintenance - hydraulic capacity - (Not asset maintenance)

- Spill frequency primarily caused by hydraulic capacity issue rather than something that can be fixed operationally
- This indicates there is insufficient capacity (conveyance or storage) in the sewer network to cope with the wastewater flow plus typical rainfall entering the sewer network.
- Also includes where currently no clear evidence that the majority of spills were due to asset maintenance issues / assets where verified hydraulic model shows >40 spills are due to hydraulic overload / sites already part of ongoing SOAF investigations

N/A - Ongoing investigation – (Not asset maintenance)

- When investigation into high spill frequency has not yet indicated the suspected primary reason by the time of the regulatory EDM Annual Return submission at the end of February.

High spill threshold not met

- The long-term average EDM Storm Overflow Annual Return high spill thresholds of greater than 50 spills if two years of data or greater than 40 spills if three or more years of data have not been exceeded.

Improvement solution delivered in previous reporting year

- Long-term average is still above reporting threshold however an improvement has been delivered to address the previously reported issue.
- WaSC expects long-term average to fall across future years.
- Should typically be accompanied by a response of ‘High spill threshold not met’ in column W (if applicable).

Column Y: Investigation activity for the reporting period

No investigation activity in reporting period

- Indicates no investigation activity took place during the calendar reporting year.

Env Act (SODRP) investigation completed

- Environment Act (Storm Overflow Discharge Reduction Plan) investigation. e.g. PR24 EnvAct_INV4. Use when the investigation work has concluded, an outcome of the investigation has been determined and/or the investigation has been signed off (if applicable).

Env Act (SODRP) investigation ongoing

- Environment Act (Storm Overflow Discharge Reduction Plan) investigation. e.g. PR24 EnvAct_INV4. Use when the investigation work has not concluded, a recommended course of action has not been determined, or the investigation has not been signed off (if applicable).

UWWTR investigation completed

- Urban Waste Water Treatment Regulations (UWWTR) investigation, e.g. U_INV. Use when the investigation work has concluded, an outcome of the investigation has been determined and/or the investigation has been signed off (if applicable).

UWWTR investigation ongoing

- Urban Waste Water Treatment Regulations (UWWTR) investigation, e.g. U_INV. Use when the investigation work has not concluded, a recommended course of action has not been determined or the investigation has not been signed off (if applicable).

Operational investigation completed

- Operational investigation activity directly linked to spill frequency or operation of the asset. Should not include investigations related to routine maintenance activity that should be completed by a WaSC. Use when the investigation work has concluded, an outcome of the investigation has been determined and/or the investigation has been signed off (if applicable).

Operational investigation ongoing

- Operational investigation activity directly linked to spill frequency or operation of the asset. Should not include investigations related to routine maintenance activity that should be completed by a WaSC. Use when the investigation work has not concluded, a recommended course of action has not been determined, or the investigation has not been signed off (if applicable).

Bathing Water and/or Shellfish Water investigation completed

- Bathing Water and/or Shellfish Water improvement activity directly related to spill frequency or impact. e.g. BW_INV. Use when the investigation work has concluded, an outcome of the investigation has been determined and/or the investigation has been signed off (if applicable)

Bathing Water and/or Shellfish Water investigation ongoing

- Bathing Water and/or Shellfish Water improvement activity directly related to spill frequency or impact. e.g. BW_INV. Use when the investigation work has not concluded, a recommended course of action has not been determined, or the investigation has not been signed off (if applicable)

Further combinations of the above options

- A maximum of one option from each of the four categories is allowed, up to a maximum of four options per overflow.

Column Z: Improvement activity for the reporting period

No improvement activity in reporting period

- Indicates no improvement activity took place during the calendar reporting

Env Act (SODRP) improvement completed

- Environment Act (Storm Overflow Discharge Reduction Plan) improvement. e.g. PR24 EnvAct_IMP4. Use when the improvement work has concluded, an outcome of the improvement has been determined and/or the improvement has been signed off (if applicable).

Env Act (SODRP) improvement ongoing

- Environment Act (Storm Overflow Discharge Reduction Plan) improvement. e.g. PR24 EnvAct_IMP4. Use when the improvement work has not concluded, a recommended course of action has not been determined, or the improvement has not been signed off (if applicable).

UWWTR improvement completed

- Urban Waste Water Treatment Regulations (UWWTR) improvement e.g. U_IMP4. Use when the improvement work has concluded, an outcome of the improvement has been determined and/or the improvement has been signed off (if applicable).

UWWTR improvement ongoing

- Urban Waste Water Treatment Regulations (UWWTR) improvement. e.g. U_IMP4. Use when the improvement work has not concluded, a recommended course of action has not been determined, or the improvement has not been signed off (if applicable).

Operational improvement completed

- Operational improvement activity directly linked to spill frequency or operation of the asset. Should not include improvements related to routine maintenance activity that should be completed by a WaSC. Use when the improvement work has concluded, an outcome of the improvement has been determined and/or the improvement has been signed off (if applicable).

Operational improvement ongoing

- Operational improvement activity directly linked to spill frequency or impact. Should not include improvements related to routine maintenance activity that should be completed by a WaSC. Use when the improvement work has not concluded, a recommended course of action has not been determined, or the improvement has not been signed off (if applicable).

Bathing Water and/or Shellfish Water improvement completed

- Bathing Water and/or Shellfish Water improvement activity directly related to spill frequency or impact. e.g. BW_IMP3. Use when the improvement work has concluded, an outcome of the improvement has been determined and/or the improvement has been signed off (if applicable).

Bathing Water and/or Shellfish Water improvement ongoing

- Bathing Water and/or Shellfish Water improvement activity directly related to spill frequency or impact. e.g. BW_IMP3. Use when the improvement work has not concluded, a recommended course of action has not been determined, or the improvement has not been signed off (if applicable).

Further combinations of the above options

- A maximum of one option from each of the four categories is allowed up to a maximum of four options per overflow.

Column AA: UWWTR (SOAF) investigation status

No SOAF investigation required

- No SOAF investigation threshold has been exceeded.

SOAF investigation newly triggered

- SOAF investigation has been triggered as a result of the reporting year spill count. Either as an individual year or contribution to the long-term average.

SOAF investigation ongoing

- SOAF investigation was triggered as a result of a previous reporting years spill count and has not yet concluded. All investigations currently in stages 1-3 which have not yet produced a decision in stage 4.

SOAF investigation completed - operational solution required

- SOAF investigation was triggered and identified an operational solution was required at stage 1b which has not yet been completed.

SOAF investigation completed - operational solution completed

- SOAF investigation was triggered and identified an operational solution was required at stage 1b which has been completed.

SOAF investigation completed - No work required

- SOAF investigation which was triggered as part of a previous reporting years spill count and has concluded that no improvement work is required. All investigations which have a decision as part of stage 4 for which the outcome is do nothing.

SOAF investigation completed - Work required

- SOAF investigation which was triggered as part of a previous reporting years spill count and has concluded that improvement work is required. Investigations which have progressed to stage 5 of the SOAF, but work has not yet concluded.

SOAF investigation completed - Work completed

- SOAF investigation which was triggered as part of a previous reporting years spill count and concluded that improvement work was required which has now been completed. Investigations which have progressed to stage 5 of the SOAF, and work has been completed.

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